

```
import re # regular expression (UNIX concept)
```

✓ Lowercasing

```
text = "Hello World!"
lowercased_text = text.lower()
print(lowercased_text)
```

```
hello world!
```

✓ Punctuation & Special Character

```
text = "Hello, world! This is?* ♥an&|~^+%" example- of text preprocessing"
text
```

```
'Hello, world! This is?* ♥an&|~^+%" example- of text preprocessing'
```

```
punctuation_pattern = r'[^\\w\\s]'
```

```
text_cleaned = re.sub(punctuation_pattern, '', text)
```

```
print(text_cleaned)
```

```
Hello world This is an example of text preprocessing
```

✓ Stop-Words

```
pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.2)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```

```
pip install stopwords
```

```
Requirement already satisfied: stopwords in /usr/local/lib/python3.12/dist-packages (1.0.2)
```

```
import nltk
```

```
nltk.download("stopwords")
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
True
```

```
from nltk.corpus import stopwords
```

```
stopwords.words("english")
```

[Show hidden output](#)

```
def remove_stopwords(text, language):
    stop_words = set(stopwords.words(language))
    text = text.lower()
    words_token = text.split()
    filtered_text = [word for word in words_token if word.lower() not in stop_words]
    print(language)
    print(filtered_text)
```

```
en_text = "This is a sample sentence and we are going to remove the stopwords."
remove_stopwords(en_text, "english")
```

```
english
['sample', 'sentence', 'going', 'remove', 'stopwords.']
```

✓ URLS

```
def remove_urls(text):
    url_pattern = re.compile(r"https?://\S+|www\.\S+")
    return url_pattern.sub(r"", text)
```

```
text = "I hope it will be a useful article for you. Follow me: https://abc.com/@def"
remove_urls(text)
```

```
'I hope it will be a useful article for you. Follow me: '
```

```
re.compile(r"https?://\S+|www\.\S+")
```

```
re.compile(r'https?://\S+|www\.\S+', re.UNICODE)
```

✓ HTML Tags

```
text = """<html><div>
<h1> Natural Language Processing</h1>
<p> Text Preprocessing for NLP</p>
<a href="https://abc.com/@def"> ABC account </a>
</div></html>"""
```

```
html_tags_pattern = re.compile(r"<.*?>")
```

```
text_without_html_tags = re.sub(html_tags_pattern, "", text)
print(text_without_html_tags)
```

```
Natural Language Processing
Text Preprocessing for NLP
ABC account
```

Start coding or [generate](#) with AI.